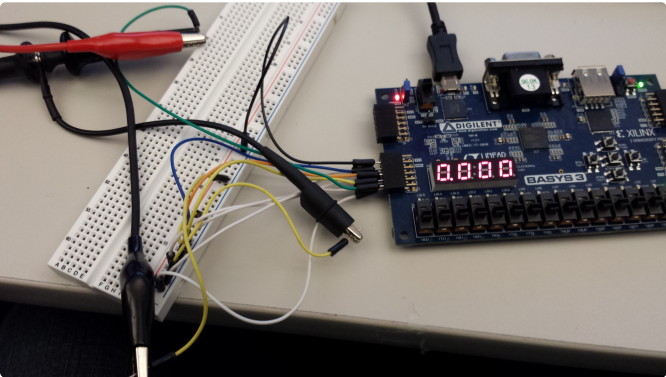


Basys 3 XADC Demo



(https://digilent.com/reference/_detail/basys3/xadc4.jpg?id=programmable-logic%3Abasys-3%3Ademos%3Axadc)

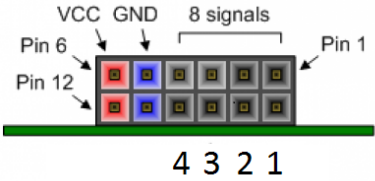
Description

This simple XADC Demo project demonstrates a simple usage of the Basys3's XADC port capability. The behavior is as follows:

- Voltage levels between 0 and 1 Volt are read off of the JXADC header.
- The 16 User LEDs increment from right to left as the voltage difference on the selected XADC pins gets larger.
- The two 7-segment displays show the voltage difference on the AD6, AD7, AD14, AD15 pins in volts.
- SW0 and SW1 select which XADC channel is displayed.

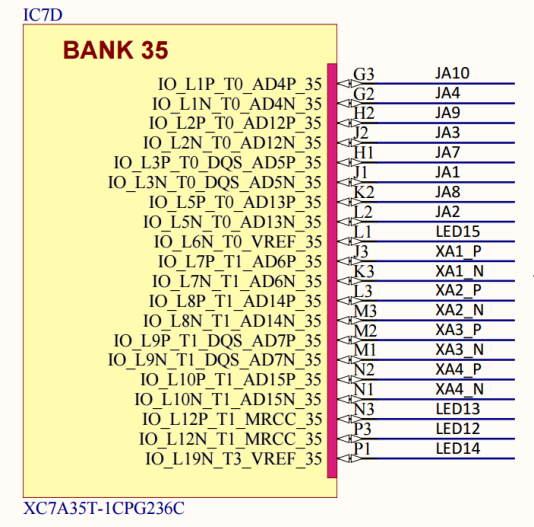
A Note About The XADC Port

On the Basys3, the XADC Pmod connector houses 4 differential analog pairs. The corresponding XADC channels are 6, 7, 14, and 15. Below is a picture of the pinout of the Pmod Header.



(https://digilent.com/reference/_detail/basys3/xadcpinout1.png?id=programmable-logic%3Abasys-3%3Ademos%3Axadc)

The differential pairs are grouped by columns. Looking at the number above, 1 corresponds to XA1_P and XA1_N, column 2 routes to XA2_P and XA2_N, 3 routes to XA3_P and XA3_N, and 4 routes to XA4_P and XA4_N. Below is from the Basys3 schematics that will show the internal connections of these signals.



(https://digilent.com/reference/_detail/basys3/basys3xadcinternal.png?id=programmable-logic%3Abasys-3%3Ademos%3Axadc)

From the schematic we can see XA1 is connected to XADC channel 6, XA2 is connected to XADC channel 14, XA3 is connected to XADC channel 7, and XA4 is connected to XADC channel 15.

Inventory

- Basys 3 with a MicroUSB Programming Cable
- Vivado installation compatible with the latest release of this demo (2022.1)
 - See *Installing Vivado, Vitis, and Digilent Board Files* (<https://digilent.com/reference/programmable-logic/guides/installing-vivado-and-vitis>) for installation instructions.
- Wires and a circuit to measure

Download and Usage Instructions

First and foremost, releases - consisting of a set of files for download - are only compatible with a specific version of the Xilinx tools, as specified in the name of the release (referred to as a *release tag*). In addition, releases are only compatible with the specified variant of the board. For example, a release tagged “20/DMA/2020.1” for the Zybo Z7 is only to be used with the -20 variant of the board and Vivado 2020.1.

The latest release version for this demo is highlighted in green.

Note: Releases for FPGA demos from before 2020.1 used a different git structure, and used a different release tag naming scheme.

Release Tag	Release Downloads	Setup Instructions
XADC/2022.1-1	Basys-3-XADC-hw.xpr.zip (https://digilent.com/reference/lib/exe/fetch.php?tok=c53560&media=https%3A%2F%2Fgithub.com%2FDigilent%2FBasys-3%2Freleases%2Fdownload%2FXADC%2F2022.1-1%2FBasys-3-XADC-hw.xpr.zip)	See <i>Using the Latest Release</i> , below
XADC/2021.1-1	Basys-3-XADC-hw.xpr.zip (https://digilent.com/reference/lib/exe/fetch.php?tok=61f924&media=https%3A%2F%2Fgithub.com%2FDigilent%2FBasys-3%2Freleases%2Fdownload%2FXADC%2F2021.1-1%2FBasys-3-XADC-hw.xpr.zip)	See <i>Using the Latest Release</i> , below
XADC/2020.1-2	Basys-3-XADC-hw.xpr.zip (https://digilent.com/reference/lib/exe/fetch.php?tok=00073d&media=https%3A%2F%2Fgithub.com%2FDigilent%2FBasys-3%2Freleases%2Fdownload%2FXADC%2F2020.1-2%2FBasys-3-XADC-hw.xpr.zip)	See <i>Using the Latest Release</i> , below
v2018.2-3	 Release ZIP downloads (https://github.com/Digilent/Basys-3-XADC/releases/tag/v2018.2-3)	 v2018.2-3 Github README (https://github.com/Digilent/Basys-3-XADC/tree/v2018.2-3)

Note for Advanced Users: *GitHub sources for this demo can be found in the  [XADC/master](https://github.com/digilent/Basys-3/tree/XADC/master) (<https://github.com/digilent/Basys-3/tree/XADC/master>) branch of the Basys-3 repository. Further documentation on the structure of this repository can be found on this wiki's [Digilent FPGA Demo Git Repositories](https://digilent.com/reference/programmable-logic/documents/git) (<https://digilent.com/reference/programmable-logic/documents/git>) page.*

Instructions on the use of the latest release can be found in this dropdown:

Using the Latest Release

Note: *This workflow is common across many Digilent FPGA demos. Screenshots may not match the demo you are working with.*

Important: *These steps are only to be used with releases for Xilinx tools versions 2020.1 and newer. Older releases may require other flows, as noted in the table of releases.*

First, download and extract the '*.xpr.zip' file from the demo release, linked above.

Open a Vivado Project from a Release

Launch Vivado

Select the dropdown corresponding to your operating system, below.

Windows

Open Vivado through the start menu or desktop shortcut created during the installation process.

(https://digilent.com/reference/_detail/learn/programmable-logic/tutorials/2020.1/launch-vivado/open-vivado.png?id=programmable-logic%3Abasys-3%3Ademos%3Axadc)

Linux

Open a terminal, and change directory (cd) to a folder where log files for your Vivado session can be placed, then run the following commands:

(https://digilent.com/reference/_detail/vivado/getting_started/linux_start_vivado.png?id=programmable-logic%3Abasys-3%3Ademos%3Axadc)

```
source <install_path>/Vivado/<version>/settings64.sh
vivado
```

In Vivado's welcome screen, use the **Open Project** button to navigate to and open the XPR file contained in the folder the release was extracted into.

(https://digilent.com/reference/_detail/learn/programmable-logic/tutorials/2020.1/using-releases/choose-xpr.png?id=programmable-logic%3Abasys-3%3Ademos%3Axadc)

Build a Vivado Project

Note that if your project already has a generated bitstream, as indicated by the status in the top right corner of the window reading “write_bitstream Complete!”, then you can skip this section.

(https://digilent.com/reference/_detail/programmable-logic/guides/write-bitstream-complete.png?id=programmable-logic%3Abasys-3%3Ademos%3Axadc)

Generate a Bitstream

In order to create a file that can be used to program the target board, each stage of the “compilation pipeline” needs to be run.

This starts with *Synthesis*. Synthesis creates a description of the logic gates and connections between them required to perform the functionality described by the HDL files, given the constraints included in XDC files. To run Synthesis click either in the toolbar or in the *Flow Navigator*. The output of Synthesis is then passed to Implementation.

Implementation has several steps. The steps that are always run are *Opt Design* (Optimize the design to fit on the target FPGA), *Place Design* (Lay out the design in the target FPGA fabric), and *Route Design* (Route signals through the fabric). To run Implementation click either in the toolbar or in the *Flow Navigator*. This output is then passed on to the Bitstream Generator.

The *Bitstream Generator* generates the final output file needed for programming the FPGA. To run Bitstream Generation click either in the toolbar or in the *Flow Navigator*. With no settings changed, the generator will create a '.bit' file.

Depending on the complexity of the design, the board used, and the strength of your computer, the process of building the project can take between 5 and 60 minutes. When complete, a pop-up dialog will appear, prompting you to select one of several options. None are relevant for the purposes of this guide, so click **Cancel**. The “write_bitstream complete” status message can be seen in the top right corner of the window, indicating that the demo is ready to be deployed to your board.

Set up the Basys 3

Plug the Basys 3 into the computer using the microUSB cable. The circuit and wires can be attached after the demo is already running.

Program a Bitstream onto an FPGA Board

Vivado's Hardware Manager can be opened by clicking on **Open Hardware Manager** at the bottom of the Flow Navigator pane on the left side of the Vivado window.

(https://digilent.com/reference/_media/vivado/getting_started/v2016.4/hardware-manager/hardware-manager.png)

The first step to programming a device is to connect the Vivado Hardware Server to it as a target. To get to the *Open Hardware Target* wizard click the link in the green banner near the top of the window. From the drop-down that opens, select .

(https://digilent.com/reference/_media/vivado/getting_started/open_target.png)

Once the wizard opens, click *Next*.

The next screen asks if the hardware server is local or remote. If the board is connected to the host computer choose local, if it is connected to another machine choose remote and fill in the *Host Name* and *Port* fields.

(https://digilent.com/reference/_media/vivado/getting_started/open_target_hw_srvr_settings.png)

Click *Next* to continue.

This screen gives a list of devices connected to the hardware server. If there is only one connected it will be the only device shown.

(https://digilent.com/reference/_media/vivado/getting_started/v2016.4/project-manager/open-new-hardware-target.png)

Click *Next* to continue.

The final screen shows a summary of the options selected in the wizard. Verify the information and click *Finish*. The board is now connected to the hardware server.

(https://digilent.com/reference/_media/vivado/getting_started/open_target_summary.png)

To program the device with the bit file generated earlier, either click the link in the *green banner* at the top of the window or click the button in the *Flow Navigator* under . From the drop-down that opens, select the device to program (Example:) and the following window will open:

(https://digilent.com/reference/_media/vivado/getting_started/v2016.4/project-manager/program-device.png)

The *Bitstream File* field should be automatically filled in with the bit file generated earlier. If not, click the button at the right end of the field and navigate to *<Project Directory>/<Project Name>.runs/impl_1/* and select the bit file (Example:). Now click *Program*. This will connect to the board, clear the current configuration, and program it using the new bit file.

At this point, the demo is now running on your board. Refer to the *Description* and *Functionality* sections of this document for more information on what it does.

Functionality

1. Applying a voltage to the XADC port

For this demo, the AD6, 7, 14 and 15 pins are used on the JXADC header. We hooked up a signal generator to our pins. All of the other pins were grounded to avoid coupling.



2. Seven Segment Display and LEDs

The 7-Segment display shows the current voltage across the selected XADC pins. The LEDs turn on from right to left as the input voltage increases.



3. Selecting a Channel

To display a different channel on the 7-segment display and LEDs change SW0 and SW1 to select the desired channel as seen in the table below.

Channel Number	SW0 / SW1 Positions
XA1	Down / Down
XA2	Up / Down
XA3	Down / Up
XA4	Up / Up

Additional Resources

All materials related to the use of the Basys 3 can be found on its [Resource Center](https://digilent.com/reference/programmable-logic/basys-3/start) (<https://digilent.com/reference/programmable-logic/basys-3/start>).

For a walkthrough of the process of creating a simple HDL project in Vivado, see [Getting Started with Hardware Design in Vivado: Essential Setup & Navigation](https://digilent.com/reference/programmable-logic/guides/getting-started-with-vivado) (<https://digilent.com/reference/programmable-logic/guides/getting-started-with-vivado>). Information on important parts of the [GUI](#), and indirect discussion of the steps required to modify, rebuild, and run this demo in hardware can also be found [here](#).

For technical support, please visit the [FPGA](https://forum.digilent.com/forum/4-fpga/) (<https://forum.digilent.com/forum/4-fpga/>) section of the Digilent Forum.

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